EmpTrack Pro

Intelligent Employee Lifecycle, Probation & Analytics System

Database Lab Project

Project Team

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1. Database Schema

Tables_in_emptrackpro
activitylog
department
employee
employment
jobrole
probationrecord
salaryhistory
systemtable
team
user

1.1 Employee Table

Description: Stores basic employee personal information.

Attributes:

- employee_id (PK)
- prefix
- first_name
- last_name
- full_name

Field	Type	Null	Key	Default	Extra
employee_id	varchar(10)	NO	PRI	NULL	
prefix	varchar(10)	YES		NULL	
first_name	varchar(50)	YES		NULL	
last_name	varchar(50)	YES		NULL	
full_name	varchar(100)	YES		NULL	

1.2 Department Table

Description: Stores department details.

Attributes:

- department_id (PK)
- department_name

Field	Type	Null	Key	Default	Extra
department_id	int	NO	PRI	NULL	auto_increment
department_name	varchar(100)	YES	UNI	NULL	

1.3 Team Table

Description: Stores teams under departments.

Attributes:

- team_id (PK)
- team_name
- department_id (FK)

Field	Type	Null	Key	Default	Extra
team_id	int	NO	PRI	NULL	auto_increment
team_name	varchar(100)	YES		NULL	
department_id	int	YES	MUL	NULL	

1.4 JobRole Table

Description: Stores job roles of employees.

Attributes:

- role_id (PK)
- role_name

Field	Type	Null	Key	Default	Extra
role_id	int	NO	PRI	NULL	auto_increment
role_name	varchar(100)	YES	UNI	NULL	

1.5 System Table

Description: Stores systems used (EMIS, SystemOne, etc.).

Attributes:

- system_id (PK)
- system_name

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Field	Type	Null	Key	Default	Extra
system_id	int	NO	PRI	NULL	auto_increment
system_name	varchar(100)	YES	UNI	NULL	

1.6 Employment Table

Description: Stores job-related details of employees.

Attributes:

- employment_id (PK)
- employee_id (FK)
- role_id (FK)
- department_id (FK)
- team_id (FK)
- system_id (FK)
- salary
- job_start_date
- exit_date
- job_status
- employment_type

Field	Type	Null	Key	Default	Extra
employment_id	int	NO	PRI	NULL	auto_increment
employee_id	varchar(10)	YES	MUL	NULL	
role_id	int	YES	MUL	NULL	
department_id	int	YES	MUL	NULL	
team_id	int	YES	MUL	NULL	
system_id	int	YES	MUL	NULL	
salary	decimal(10,2)	YES		NULL	
job_start_date	date	YES		NULL	
exit_date	date	YES		NULL	
job_status	varchar(50)	YES		NULL	
employment_type	varchar(50)	YES		NULL	

1.7 ProbationRecord Table

Description: Tracks employee probation period.

Attributes:

- probation_id (PK)
- employee_id (FK)
- probation_start
- probation_end
- status

Field	Type	Null	Key	Default	Extra
probation_id	int	NO	PRI	NULL	auto_increment
employee_id	varchar(10)	YES	UNI	NULL	
probation_start	date	YES		NULL	
probation_end	date	YES		NULL	
status	varchar(50)	YES		NULL	

1.8 User Table

Description: Stores login and role-based access information.

Attributes:

- user_id (PK)
- username
- password
- role

Field	Type	Null	Key	Default	Extra
user_id	int	NO	PRI	NULL	auto_increment
username	varchar(50)	YES	UNI	NULL	
password	varchar(100)	YES		NULL	
role	varchar(50)	YES		NULL	

1.9 Activity Log

Description: Stores records of all actions performed by employees or users in the system for tracking and monitoring.

Attributes:

- log_id (PK)
- employee_id(FK)
- action
- action_time

Field	Type	Null	Key	Default	Extra
log_id	int	NO	PRI	NULL	auto_increment
employee_id	varchar(20)	YES	MUL	NULL	
action	varchar(100)	YES		NULL	
action_time	datetime	YES		NULL	

1.10 Salary History

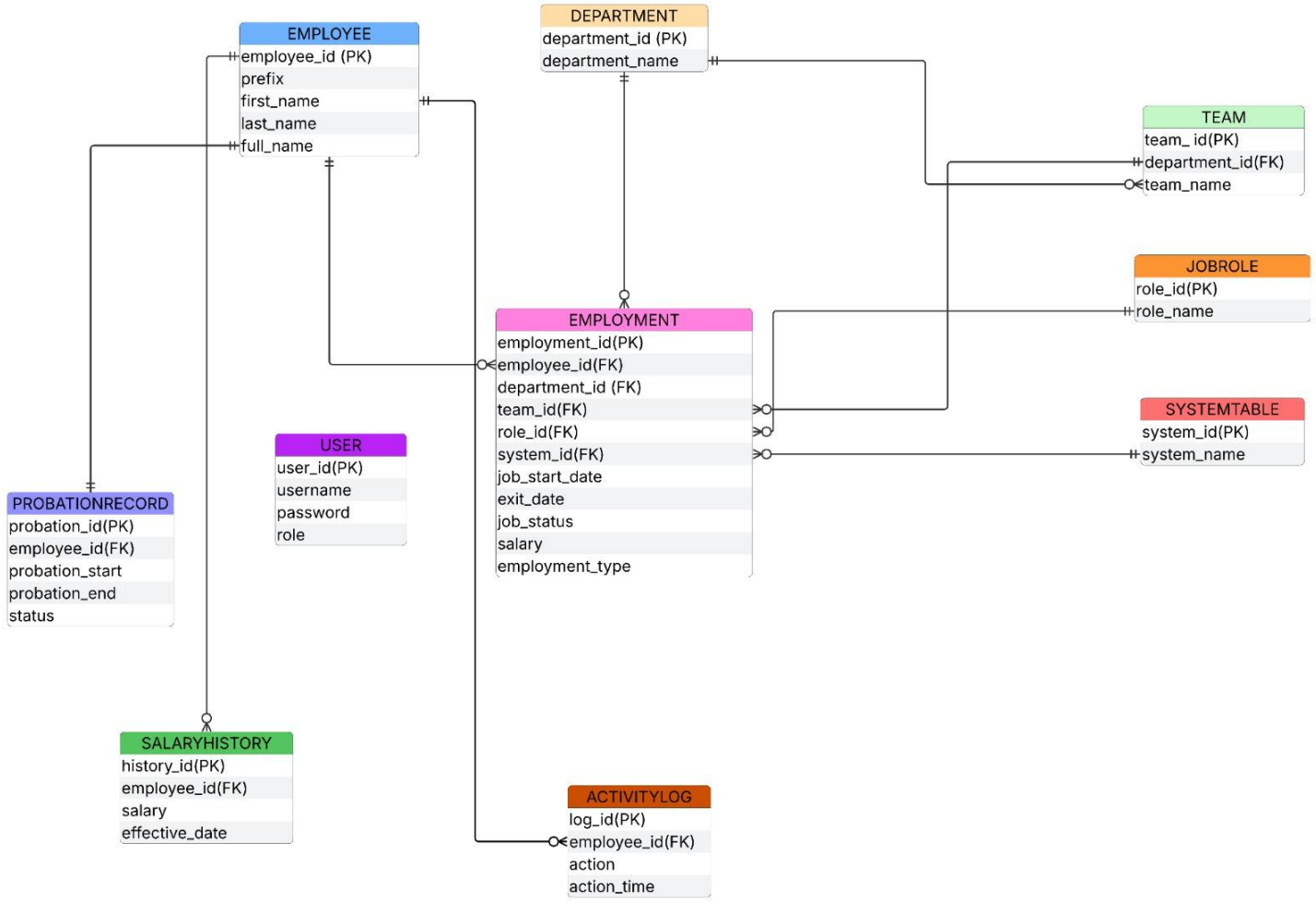
Description: Maintains a record of changes in an employee's salary over time.

Attributes:

- history_id (PK)
- employee_id (FK)
- salary
- effective_data

Field	Type	Null	Key	Default	Extra
history_id	int	NO	PRI	NULL	auto_increment
employee_id	varchar(20)	YES	MUL	NULL	
salary	decimal(10,2)	YES		NULL	
effective_date	date	YES		NULL	

2.Relationships Between Entities

- **Department → Team**
  One department can have multiple teams.
  (1-to-Many)
- **Department → Employment**
  One department can have multiple employees (via employment records).
  (1-to-Many)
- **Team → Employment**
  One team can have multiple employees.
  (1-to-Many)
- **Job Role → Employment**
  One job role can be assigned to multiple employees.
  (1-to-Many)
- **System → Employment**
  One system can be used by multiple employees.
  (1-to-Many)
- **Employee → Employment**
  One employee can have multiple employment records (job history).
  (1-to-Many)
- **Employee → Probation Record**
  One employee has one probation record.
  (1-to-One)
- **Employee → Activity Log**
  One employee can have multiple activity logs.
  (1-to-Many)
- **Employee → Salary History**
  One employee can have multiple salary history records.
  (1-to-Many)
- **Employee → User**
  One employee is associated with one system user account.
  (1-to-One)

3.Entity Relationship Diagram (ERD)